AIA SPARK

High-level architecture

Voice-first debrief, agentic product suggestions and campaigns, with all components inside one Azure subscription

Azure subscription {BU}-01

Advisor

Web browser on desktop or mobile, with microphone and speaker for the voice debrief

HTTPS

Azure subscription {BU}-01

Azure Key Vault

Secrets for both pods: model keys, database and Redis credentials, integration credentials. Never stored in images or config files.

secrets for both pods: UI and API

AZURE KUBERNETES SERVICE (AKS)

Ingress · HTTPS · REST · WebSocket (voice) · Server-sent events (live analysis)

Spark UI pod

REACT · NGINX

- Voice debrief with Spark
- Suggestions, proposal, advice pack
- Campaigns on the suggestion page
- Admin: knowledge, business unit, campaigns

Spark API pod

SPRING BOOT · JAVA 21

- Voice sessions and live transcription
- Spark conversation and spoken replies
- Agent graph (LangGraph4j): 11 analysis agents
- Campaign agent, run beside the graph
- Advice pack, proposal and reports
- Settings, sessions and rate limiting

SQL + vector search

sessions · limits

Azure Database for PostgreSQL

Flexible Server · pgvector extension enabled

Application database

Customer profiles, debriefs, runs, advice packs, campaigns, users, settings

pgvector database

Product-document chunks and campaign embeddings (3072 dimensions)

Azure Cache for Redis

Session tokens and request rate limiting

chat · vision · speech · embeddings

AZURE OPENAI IN AZURE AI FOUNDRY four model deployments, in our subscription

gpt-4.1

CHAT · VISION

Reasoning model behind Spark and the analysis agents. Extracts the fact-find, writes suggestions and proposals, and reads campaign posters.

gpt-4o-transcribe

SPEECH TO TEXT

Turns the advisor's live speech into text, across English, Mandarin, Malay, Tamil, Thai, Filipino and Vietnamese, including mixed languages.

gpt-4o-mini-tts

TEXT TO SPEECH

Gives Spark its natural voice (Coral and Ash) in every supported language, with a tone that fits each line.

text-embedding-3-large

EMBEDDINGS

Turns text into 3072-dimension vectors to search product documents and to match a customer's words to campaigns.

Integrations

Enterprise and national systems

iPOS

SPARK → iPOS

Spark sends the FHR report once the debrief and suggestions are complete.

CRM

SPARK → CRM

Spark passes the follow-up tasks from the meeting to the CRM.

i-Smart

i-SMART → SPARK

Spark retrieves customer information for the advisor's own customers.

ODS (Operational Data Store)

ODS → SPARK

Spark retrieves the customer's existing AIA policy information.

SGFinDex

SGFinDex → SPARK

Spark retrieves the customer's policies held with other insurers, with their consent.

Solid red arrows: runtime calls inside the subscription. Dashed: secrets. Dark arrows: integrations with enterprise and national systems.